

B. Tech 4th SEM (Mechanical Engineering)
Strength of Material-II (2nd Sessional Test)

Time: 1.5 hours

Maximum Marks: 15

Note: 1) All questions are compulsory. Each question is of 5 Marks. Assume the missing data if any.

Q1. Prove that the thickness of disc of uniform strength is given by

$$t = t_0 e^{-\rho\omega^2 r^2 / 2\sigma}$$

Where t_0 = thickness at $r_0 = 0$

σ_0 = stress developed due to rotation.

(CO3)

Q2. A bending moment of 4 kN is found to cause elastic failure of a solid circular shaft. An exactly similar shaft is now subjected to a torque T. Determine the value of T which causes failure of the shaft according to the following theories:

(i) Maximum principal stress

(ii) Maximum principal strain

(iii) Maximum Strain Energy Theory (Poisson ratio = 0.3)

(CO4)

Q3. A T-section 150 mm × 120 mm × 20 mm is used as a strut of 4 m long with hinged at its both ends. Calculate the crippling load, if Young's modulus for the material be 200 GPa. (CO5)